

Zijian (Jack) Luo

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Education

Monash University <i>PhD in Computer Science</i>	Incoming <i>Australia</i>
The University of Sydney <i>MPhil in Computer Science</i>	Mar 2024 – Mar 2026 <i>Sydney, Australia</i>
Hebei University <i>B.Eng. in Software Engineering</i>	Sep 2018 – Jun 2022 <i>Hebei, China</i>

Work Experience

-  **Microsoft Research Asia (MSRA), Microsoft** Mar 2026 – Present
Research Intern
- Enhanced **Copilot's cross-platform code migration** capability by integrating a **knowledge base** to model platform-specific APIs, dependencies, and migration mappings.
 - Built a **dependency-upgrade benchmark** for **long-horizon software engineering tasks** to evaluate **MAI Coder's** end-to-end planning and execution across multi-step workflows; the corresponding paper is under review.
 - Mined **hidden knowledge** from codebases to provide project-specific context for **Copilot Code Review (CCR)**.
-  **Oracle Labs, Oracle** May 2025 – Oct 2025
Research Assistant
- Conducted research on **software supply-chain reliability** by integrating LLMs with the Macaron framework, reproducing build artifacts for over **120 open-source repositories** from Libraries.io with an **80% success rate**.
 - Designed LLM-assisted workflows for defect reproduction and **static-analysis** validation.

Publications

- Automatic Data Repair without Format Specifications.**
Zijian Luo, Lukas Kirschner, Ezekiel Soremekun, Rahul Gopinath.
IEEE International Symposium on Software Reliability Engineering (ISSRE) 2025. CCF B CORE A
- Assessing Reliability of Statistical Maximum Coverage Estimators in Fuzzing.**
Danushka Liyanage*, Nelum Attanayake*, **Zijian Luo***, Rahul Gopinath.
IEEE International Conference on Software Maintenance and Evolution (ICSME) 2025. CCF B CORE A
- Maximal Format-Free Record Repair.**
Zijian Luo, Xi Wu, Hong Jin Kang, Alan Fekete, Rahul Gopinath.
IEEE/ACM International Conference on Automated Software Engineering (ASE) 2026. CCF A CORE A*

Selected Projects

- The Deeplang Programming Language (82 Stars)**
A memory-safe, multi-paradigm programming language for embedded systems. I was responsible for implementing the backend virtual machine, including interpreting and executing WebAssembly (WASM) code generated by the frontend.

Technical Skills

Programming Languages	C/C++, Python, Rust, WebAssembly
Analysis & Testing	Fuzzing (AFL), Program Repair, Grammar Inference, CodeQL
AI	LLMs